Properties of z-Transform

* Time reversal

$$\{x(n)\} \xleftrightarrow{z} X(z) \quad \text{with ROC} \quad r_1 < |z| < r_2$$

$$x_1(n) = x(-n)$$

$$\{x_1(n)\} \xleftrightarrow{z} X_1(z) = X(1/z)$$

ROC: $r_1 < \frac{1}{|z|} < r_2$, i.e. $\frac{1}{r_2} < |z| < \frac{1}{r_1}$

$X_1(z)$ = z transform of $\{x_1(n)\}$ at point z

$X(1/z)$ = z-transform of $\{x(n)\}$ evaluated @ $1/z$

Proof:

$$x_1(n) = x(-n)$$

$$X_1(z) = \sum_{n=-\infty}^{\infty} x_1(n) z^{-n}$$

$$= \sum_{n} x(-n) z^{-n}$$

$$l = -n \quad = \sum_{l=-\infty}^{\infty} x(l) z^{l}$$

$$= \sum_{l=-\infty}^{\infty} x(l) \left(\tfrac{1}{z}\right)^{-l}$$

$$X_1(z) = X(1/z)$$

$1/z$ should lie in ROC of $X(z)$

If $X(z)$ has a pole (zero) @ $z = \alpha$

then $X_1(z)$ will have a pole (zero) @ $z = 1/\alpha$

$r_1 < \frac{1}{|z|} < r_2$

$\frac{1}{r_2} < |z| < \frac{1}{r_1}$

Property

Conjugation $\{x(n)\} \xleftrightarrow{Z} X(z)$ with ROC R

$\{x^*(n)\} \xleftrightarrow{Z} X^*(z^*)$ with ROC R

$\{x^*(n)\} = \{x_1(n)\}$

$X^*(z^*) = X_1(z)$

$X_1(z)$ = $X^*(z^*)$

z transform of $\{x_1(n)\}$ evaluated at point z

Conjugate of (z transform of $\{x(n)\}$ evaluated @ z^*)

Since z & z^* has same magnitude.

$X_1(z)$ is well defined (ROC) if $X(z^*)$ is well defined (ROC)

Symmetry Real signal

$x(n) = x^*(n) \quad \forall n$

real signal $\downarrow z \qquad \downarrow z$

$X(z) = X^*(z^*)$ real signals

If $z = \alpha$ is a pole (zero)

then $z = \alpha^*$ is also a pole (zero)

For real valued signals, the poles and zeros of z-transform appear in conjugate pairs.

Exponential Multiplication

$$\{x(n)\} \xleftrightarrow{z} X(z) \quad ; \quad r_1 < |z| < r_2$$

$$x_1(n) = x(n)\, z_0^n \qquad n \in \mathbb{Z}$$

$$\{x_1(n)\} \xleftrightarrow{z} X_1(z)$$

$$X_1(z) = X(z/z_0) \qquad r_1 < \frac{|z|}{|z_0|} < r_2$$

Differentiation Property

$$\{x(n)\} \xleftrightarrow{z} X(z) \quad \text{with} \quad \text{ROC} \quad R$$

$X(z)$ is analytic function in ROC R (differentiable)

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}$$

$$\frac{d}{dz} X(z) = \sum_{n=-\infty}^{\infty} x(n) \frac{d}{dz} z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x(n)(-n)\, z^{-n-1}$$

$$-z \frac{d}{dz} X(z) = \sum_{n=-\infty}^{\infty} n\, x(n)\, z^{-n}$$

z transform of $\{n\, x(n)\}$

$\{x(n)\} \xleftrightarrow{z} X(z)$ with ROC R

$\{n\,x(n)\} \xleftrightarrow{z} -z\frac{d}{dz}X(z)$ with ROC R



Example:

$x(n) = a^n u(n)$ $\quad a \in \mathbb{C}$

$$X(z) = \frac{1}{1-az^{-1}} \quad ; \quad |z| > |a|$$

$\{n\,x(n)\} \xleftrightarrow{z} -z\frac{d}{dz}X(z) \quad ; \quad |z| > |a|$

$n a^n u(n) \xleftrightarrow{z} \frac{az^{-1}}{(1-az^{-1})^2} \quad ; \quad |z| > |a|$

$\Downarrow$

Timing advance by 1 unit

$(n+1)\,a^{n+1}\,u(n+1) \xleftrightarrow{z} \frac{a}{(1-az^{-1})^2} \quad ; \quad |z| > |a|$

$(n+1)\,a^n\,u(n+1) \xleftrightarrow{z} \frac{1}{(1-az^{-1})^2} \quad ; \quad |z| > |a|$

$\|$

$(n+1)\,a^n\,u(n) \xleftrightarrow{z} \frac{1}{(1-az^{-1})^2} \quad ; \quad |z| > |a|$



Convolution Property

input $\{x(n)\} \xleftrightarrow{\mathcal{Z}} X(z)$ with ROC R_x

impulse response $\{h(n)\} \xleftrightarrow{\mathcal{Z}} H(z)$ with ROC R_h

$\{y(n)\} = \{x(n)\} * \{h(n)\}$ Convolution

$\{y(n)\} \xleftrightarrow{\mathcal{Z}} Y(z) = H(z)X(z)$ with

ROC $R_y \supset R_x \cap R_h$

R_y is at least as big as $R_x \cap R_h$

Examples

1. Finite length signals

$\{x(n)\} = \{\underset{\downarrow n=0}{1}, 2, 1\}$ $\{h(n)\} = \{\underset{\downarrow n=0}{1}, -1\}$

$X(z) = 1 + 2z^{-1} + z^{-2}$ $H(z) = 1 - z^{-1}$

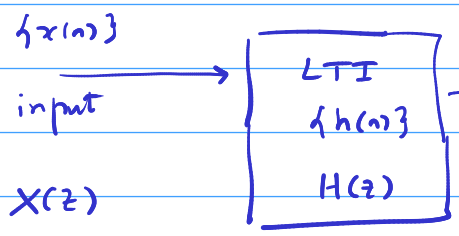
$|z| > 0$ $|z| > 0$

$Y(z) = X(z)H(z) = 1 + z^{-1} - z^{-2} - z^{-3}$

$|z| > 0$

$\{y(n)\} = \{\underset{\downarrow n=0}{1}, 1, -1, -1\}$

LTI Systems & Z-transforms



output

$\{y(n)\} = \{x(n)\} * \{h(n)\}$ — Convolution

$Y(z) = X(z) \cdot H(z)$ — Multiplication

$\{h(n)\} \xleftrightarrow{Z} H(z)$ with ROC R

impulse response

Transfer function

Another way to characterize LTI systems

Example: Infinite length convolution

$x(n) = u(n)$ $\qquad h(n) = a^n u(n)$

$|a| < 1$

(absolutely summable)

Stable LTI

$\{y(n)\} = \{x(n)\} * \{h(n)\}$

$$X(z) = \frac{1}{1 - z^{-1}} \quad ; \quad |z| > 1$$

$$H(z) = \frac{1}{1 - az^{-1}} \quad ; \quad |z| > |a|$$

$$Y(z) = X(z)\,H(z) = \frac{1}{1 - z^{-1}} \cdot \frac{1}{1 - az^{-1}} \quad ; \quad |z| > 1$$

Partial Fraction ⇒

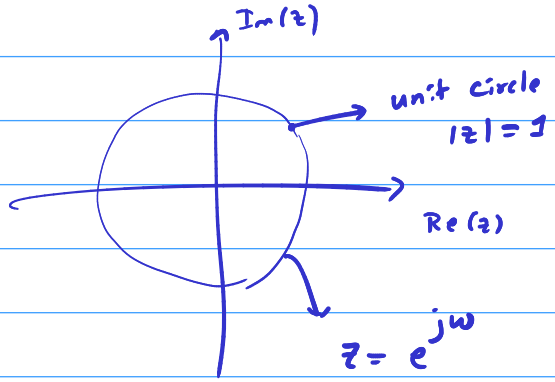
$$Y(z) = \frac{1}{(1-a)} \cdot \frac{1}{1 - z^{-1}} - \frac{a}{1-a} \cdot \frac{1}{1 - az^{-1}} \quad ; \quad |z| > 1$$

$$y(n) = \frac{1}{(1-a)}\,u(n) - \frac{a}{1-a}\,a^{n}\,u(n)$$

Evaluating Sums

$$\{x(n)\} \overset{z}{\longleftrightarrow} X(z) \quad \text{with} \quad \text{ROC} \quad R$$

Suppose ROC includes unit circle



For any $z \in$ ROC

$$z = re^{j\omega}$$

$$\sum_{n=-\infty}^{\infty} |x(n)|\, r^{-n} < \infty$$

Absolute convergence

If $|z| = 1$ is in ROC (unit circle)

$$\sum_{n=-\infty}^{\infty} |x(n)| \underbrace{1^{-n}}_{1} < \infty$$

$\Rightarrow$ $x(n)$ sequence is absolutely summable.

$$\sum_{n=-\infty}^{\infty} x(n) \quad \text{exists}$$

$$X(z)\Big|_{@\, z=1} = \sum_{n=-\infty}^{\infty} x(n)\, 1^{-n} = \sum_{n=-\infty}^{\infty} x(n)$$

$$X(z=1) = \sum_{n=-\infty}^{\infty} x(n)$$

LTI System Properties & Transfer function

LTI System impulse response $\{h(n)\}$

$\Downarrow$ z transform

Transfer function $H(z)$ with ROC R.

1. Causal LTI System

$h(n) = 0 \quad \text{for} \quad n < 0$

$$H(z) = h(0) z^{0} + h(1) z^{-1} + h(2) z^{-2} + \dots$$

$H(z) \to h(0)$ as $z \to \infty$

$|z| = \infty$ is in ROC R

For a rational transfer function causal LTI system ROC will be outside largest pole.

2. Stable LTI system

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty \quad \text{(absolutely summable)}$$

ROC of $H(z)$ containing unit circle

$|z| = 1$ should be in ROC of Transfer function for stable LTI systems.

3. Causal & stable LTI System

Both $|z| = 1$ and $|z| = \infty$ should be in ROC.

If $H(z)$ is rational transfer function

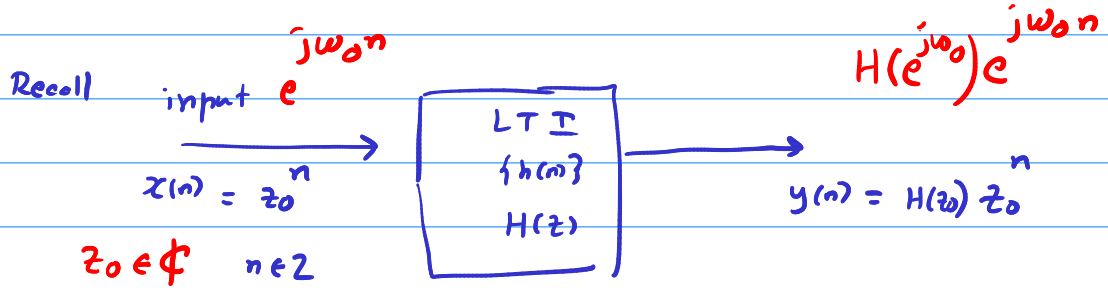
All poles should lie inside unit circle.

$\times \longrightarrow \times$

Discrete Time Fourier Transform (DTFT)

- Gives us frequency contents of signal
- Useful to study LTI systems.
- Analytical tool

Recall



$$H(z_0) = \sum_{n=-\infty}^{\infty} h(n) z_0^{-n}$$

$$= H(z)\Big|_{@\, z = z_0}$$

Special case

$z_0 = e^{j\omega_0}$ $(r = 1)$ $|z_0| = 1$

$$H(e^{j\omega_0}) = H(z)\Big|_{@\, z = e^{j\omega_0}}$$

$$= \sum_{n=-\infty}^{\infty} h(n) e^{-j\omega_0 n}$$

Discrete time Fourier transform (DTFT) of DT signal $x(n)$ is given by

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$e^{j\omega n} = e^{j(\omega+2\pi)n}$$

real variable $-\infty < \omega < \infty$

Since DT signal $e^{j\omega n} \equiv e^{j(\omega+2\pi)n}$

we can restrict

$$-\pi \leq \omega < \pi$$

$$0 \leq \omega < 2\pi$$